

Chapter 13 — Incisions, Exposure, and Drains

- **Open-surgery exposure has fallen** with the expansion of laparoscopy and robotic-assisted laparoscopy, but **open skill remains necessary**. Open versus minimally invasive is decided by **complexity and size of the pathology, patient preference and the surgeon's skill set**.

ABDOMINAL INCISIONS

Midline Incision

- **Most versatile incision:** reaches the extraperitoneal, peritoneal and retroperitoneal spaces. Either half can be carried past the umbilicus.

MIDLINE INCISION — VARIANTS

Variant	Access	Indications
Lower (infraumbilical)	Pelvis: extraperitoneal retropubic space; peritoneum if required	Radical retropubic and simple prostatectomy, pelvic lymph node dissection (PLND), radical and partial cystectomy, diverticulectomy, enterocystoplasty, urinary diversion; specimen extraction (e.g. kidney after laparoscopy)
Upper (supraumbilical)	Bilateral retroperitoneum: adrenals, kidneys, ureters, nodes, great vessels	Retroperitoneal lymph node dissection (RPLND), bilateral simple nephrectomy, retroperitoneal mass
Complete	Whole abdomen and retroperitoneum	RPLND, bilateral nephrectomy for polycystic kidneys, large retroperitoneal mass, nephroureterectomy, bowel-heavy reconstruction (Indiana pouch, ileal ureter); trauma nephrectomy

- **Incise along the linea alba** (the midline aponeurosis of the abdominal wall muscles), not through rectus abdominis. This avoids the vessels that run deep to and perforate the muscle, so it bleeds less (Fig. 13.1). It is harder in **obesity, multiple prior laparotomies and hernia**.
- **Entry:** extraperitoneal → open posterior fascia **as low as possible near the pubis** → expand bluntly. Intraperitoneal → posterior fascia → peritoneum.
- **Closure in one layer**, under more tension than incisions that follow **Langer lines**. Fascia running or interrupted figure-of-eight, strong absorbable suture (monofilament polydioxanone or braided polyglycolide). **Bites 1 cm from the edge, 1 cm apart** (Fig. 13.2).
- **No closure method is superior** for dehiscence or incisional hernia, but **avoid continuous closure with a rapidly absorbable suture:** it raises hernia rates.

OTHER FASCIAL CLOSURE TECHNIQUES (FIG. 13.3) — LESS USED IN UROLOGY

Technique	Construction
Near-and-far	Mass closure
Smead-Jones	Near and far figure-of-eight bites, sutures 2 cm apart
Gambie / vertical mattress	Incorporates both fascial layers

Source: Hinman's Atlas of Urologic Surgery, 4th edition, 2018.

Pfannenstiel Incision

- **Adult and paediatric;** extraperitoneal or intraperitoneal pelvic surgery, benign or malignant; **donor nephrectomy and post-nephrectomy specimen extraction**.
- **Lower morbidity and incisional hernia rates** than midline or port-site extension. Good cosmesis and less pain, because the rectus is **separated, not cut**.

Sequence: transverse semilunar skin incision **two fingerbreadths above the pubis** → anterior rectus sheath opened transversely → rectus bluntly freed from the sheath above and below → linea alba attachment taken down superiorly and inferiorly → midline opened → extraperitoneal space or peritoneum.

- **Extensions:** laterally for exposure; **into a Gibson** for the ureter; skewed toward the operative side for extraction.
- **Closure:** peritoneum and muscle reapproximated; fascia running or figure-of-eight absorbable.

Gibson Incision

- **Traditional incision for renal transplantation and transplant nephrectomy.** Muscle-splitting, **extraperitoneal** access to the iliac vessels, lower ureter and bladder. Also the hand-port or extraction site in minimally invasive nephrectomy.
- **Traditional oblique vs “hockey-stick” (J-shaped):** oblique runs from a few centimetres medial to the anterior superior iliac spine (ASIS) toward the inguinal fold, ending just lateral to rectus. The J can be extended further superiorly, but **retrospective series show higher hernia rates**.

Sequence: external oblique split → internal oblique → transversus abdominis (incise if needed) → **peritoneum mobilised medially** → ureter found **crossing the iliac artery**. Open the peritoneum only for a hand port. Close in **two layers**, running absorbable.

Concept — *The midline gives the widest, fastest field but is one layer under tension. The transverse and muscle-splitting incisions (Pfannenstiel, Gibson) give a smaller field for less pain and fewer hernias. Staying in avascular planes such as the linea alba is how bleeding and nerve injury are avoided.*

ANTERIOR APPROACHES TO THE KIDNEY AND RETROPERITONEUM

- **Six anterior incisions** (Fig. 13.6): midline transperitoneal, subcostal, chevron, modified thoracoabdominal, transverse, paramedian. The choice is driven by **oncological and patient factors and surgeon preference**.
- **Transverse upper abdominal:** mainly **paediatric** — Wilms tumour resection, open pyeloplasty.
- **Paramedian:** described for extraperitoneal nephroureterectomy and paediatric bladder augmentation. Lateral to rectus it **risks the nerves within rectus and ventral hernia**, so it is kept for avoiding the midline in a complex abdomen (e.g. colostomy).

TABLE 13.1 OPEN APPROACHES TO THE KIDNEY

Incision	Indications	Benefits	Limitations
Anterior			
Midline transperitoneal	Trauma, inferior vena cava (IVC) thrombus, bilateral renal/ureteral disease, horseshoe kidney	Rapid, less painful, both kidneys, early vascular control	Poor upper retroperitoneal exposure; bowel and wound complications
Subcostal	Radical nephrectomy, ureteropelvic junction obstruction (UPJO)	Extends to chevron or flank; early vascular control	Bowel complications
Chevron	Bilateral renal tumours, IVC thrombus, bilateral polycystic nephrectomy, local tumour invasion	Excellent bilateral exposure; early vascular control	Liver/spleen injury; large muscles transected
Transverse abdominal	Paediatric Wilms tumour, paediatric pyeloplasty	Easy access to pedicle and retroperitoneal nodes	Unfamiliar in adults
Paramedian	Avoiding another structure (e.g. colostomy)	Develops preperitoneal space for extraperitoneal route	Superior epigastric artery, thoracic nerves; hernia

Incision	Indications	Benefits	Limitations
Modified thoracoabdominal	Radical nephrectomy, lymphadenectomy, retroperitoneal mass	Versatile; large upper-pole or adrenal masses	Bowel and wound complications, large muscles cut, painful
Flank			
11th/12th rib supracostal	Partial nephrectomy, simple nephrectomy, simple adrenalectomy	Extraperitoneal; good renal exposure; rib not cut	Pleural injury
11th rib transcostal		Good renal and retroperitoneal exposure	Pleural injury, flank bulge
Thoracoabdominal	Large renal mass, IVC thrombus, adrenal mass, invasion of adjacent structures, lymphadenectomy	Excellent exposure; can stay fully extraperitoneal	Pleural injury, large muscles cut, bowel complications if intraperitoneal

Modified from Gregg and Scarpato, Hinman's Atlas of Urologic Surgery, 4th edition, 2018, Table 8.1.

- **What the table tests:** **anterior = early vascular control and bilateral access, at the cost of bowel complications; flank = extraperitoneal, at the cost of the pleura.** Only transcostal adds flank bulge; only supracostal spares the rib.
- **Trap:** thoracoabdominal appears in both groups — the modified (anterior) version is transperitoneal, while the flank version can be completely extraperitoneal.

Thoracoabdominal Incision

- **Indications are shrinking** with incidental detection and nephron-sparing surgery. A large, morbid incision that gives **the best exposure** of the thorax, retroperitoneum and great vessels, and so **early vascular control**.
- **Indications:** **level III caval thrombus**, T4 renal tumour, large upper-pole tumour, adrenal tumour, upper retroperitoneal mass.
- **Pain is no worse than a flank incision** over the first month — counter to surgical intuition.

Sequence: exaggerated flank position, ipsilateral arm on a Mayo stand or trapeze → incision over the **9th–10th rib bed** (supracostal or with rib resection), continued onto the abdomen and to the midline if needed → muscles divided → pleura opened, sparing lung → diaphragm opened to join the cavities → **costochondral junction divided**.

- **Modified version:** thoracotomy through the **8th rib bed** + subcostal + longitudinal abdominal incision.
- **Closure:** chest tube (may be omitted if the **parietal pleura is uninjured**) → all layers with costal cartilage reapproximated → **diaphragm in two layers**, running absorbable → abdominal wall. Nonabsorbable mesh (Marlex) with anchoring figure-of-eight sutures has been described to reduce costochondral displacement and discomfort; the author does not use it.

Chevron and Subcostal Incisions

- **Subcostal transperitoneal:** renal mass with caval thrombus. **Supine subcostal vs flank subcostal:** supine gives **better hilar control for large anterior tumours**.
- **Extensions (Fig. 13.7):** midline to xiphoid → **suprahepatic IVC**; add sternotomy → **thrombus reaching the heart** (rarely used by urologists); across the midline → “three-quarters” → **full chevron** for a left-sided caval thrombus or locally invasive tumour.

Sequence: supine or modified flank → from midline **two fingerbreadths below the costal margin** to the **tip of the 11th rib** → anterior rectus sheath incised, rectus cut with cautery, lateral muscles divided in sequence → preferably intraperitoneal → incise the **white line of Toldt** to the hepatic or splenic flexure, colon

mobilised medially → self-retaining retractor (Bookwalter), guarding liver and spleen → two-layer running absorbable closure.

Concept — Anterior approaches buy **bilateral access and early control of the pedicle and IVC** by entering the peritoneum. Each step up the scale (subcostal → chevron → thoracoabdominal → sternotomy) is chosen by how far the tumour or thrombus reaches.

FLANK INCISIONS

- **Often cut muscle and/or remove rib.** Careful technique limits pain and long-term complications, and they are **not more painful than thoracoabdominal**. In the minimally invasive era they are reserved for **large or complex oncological resections** and **simple nephrectomy for xanthogranulomatous pyelonephritis (XGP)**.
- **Three muscle/fascial layers:** external oblique → internal oblique → transversus abdominis, usually divided with cautery.

FLANK VS ANTERIOR APPROACH — THE GOVERNING CONTRAST

	Flank	Anterior
Compartment	Extraperitoneal, posterior	Usually transperitoneal
Organ at risk	Pleura (± flank bulge)	Bowel (and wound)
Reach	One kidney	Both kidneys, great vessels; early vascular control

True Flank Incisions

- **Flank position:** lateral recumbent, **lower leg flexed, upper leg extended**, operative side made convex, incision over the **table break**.
- **Options** by target and size (Fig. 13.8): 12th rib supracostal, 11th rib transcostal, subcostal (also thoracoabdominal, Foley muscle-splitting). Start anterior to the 12th rib laterally, going no further medially than the **edge of rectus**.

12th Rib Supracostal and 11th Rib Transcostal

- **More posterior access → less bowel risk; more thoracic risk (pleura).** Resecting the rib tip → **significant pain; flank bulge** if as high as the 11th rib.

Sequence: skin over rib → external oblique and latissimus dorsi divided over the rib, beyond its tip → periosteal elevator strips muscle and periosteum anteriorly and along both edges → **Doyen rib stripper** posteriorly → rib cutter → edges blunted, **bone wax** → rib bed divided → lumbodorsal fascia → **pleura identified with large anaesthetic breaths** → retroperitoneum entered **as posteriorly as possible**, peritoneum pushed medially → **neurovascular bundle between layers 2 and 3** freed → retractors → close in two layers, running absorbable.

Pleural injury: red rubber catheter through the tear → tear closed around it with running absorbable → diaphragmatic muscle closed → catheter end **under water** → sustained high-volume breaths → withdraw slowly **once bubbling stops**. Large tear → **formal chest tube**.

- **Post-operative chest radiograph only if a pleural tear was recognised and repaired** (or there is clinical suspicion). Routine films after open nephrectomy have a **very low pneumothorax yield**.
- **Rib-sparing alternative:** incise above the 11th rib or **between 11th and 12th (infra-11th, Fig. 13.9)** without excising rib. This avoids rib-resection pain and complications, still exposes kidney and hilum, and stays extraperitoneal.

Reopening a Prior Incision

- Main danger is **bowel injury on transperitoneal re-entry**; there is little evidence to guide it.
- **Principle: extend the incision into unviolated tissue and enter the peritoneum there**, where planes are anatomical and adhesions fewer. Then dissect organs stuck to the old scar under direct vision.
- **Involve a general surgeon** when heavy adhesions are expected: prior **cystectomy with urinary diversion**, abdominal trauma, bowel surgery.

Concept — *The flank trades the peritoneum for the pleura: each step higher on the rib cage improves upper-pole reach but adds pleural risk, pain and bulge. That is why rib-sparing supracostal or infra-11th variants are preferred where they give enough exposure.*

INGUINAL INCISIONS

- Give access to the **spermatic cord** in children and adults: cryptorchidism with synchronous hernia repair, **partial or radical orchiectomy**.

Incisions Above the Inguinal Ligament

INGUINAL INCISIONS (FIG. 13.10)

Indication	Incision
Radical orchiectomy (commonest adult use)	5–7 cm curvilinear or transverse, starting 2 cm cephalad and lateral to the pubic tubercle ; angle more obliquely toward the scrotum for large masses
Benign: inguinal varicocelelectomy (especially prepubertal), orchidopexy ± hernia repair	2–3 cm, two fingerbreadths above the symphysis , from the lateral scrotum obliquely along the canal; aim just below the external ring (palpable in thin patients and children)
Obesity or prior inguinal surgery	Subinguinal approach
Inguinal lymphadenectomy (penile cancer)	Subinguinal, below the ligament

- **Protect the ilioinguinal nerve** and cord structures.
- **Closure:** reapproximate internal and external oblique to close the canal and form a new external ring — **not too tight**, or the cord is compromised.

Incisions for Inguinal and/or Pelvic Lymphadenectomy in Penile Cancer

- **Extent of nodal disease or planned dissection decides the incision.**

INCISIONS FOR INGUINAL (ILND) AND PELVIC (PLND) LYMPH NODE DISSECTION (FIG. 13.11)

Incision	Use
Subinguinal, parallel to ligament	Commonest ; ipsilateral nodes; modified or complete ILND
Vertical	Superficial and deep ILND
S-shaped	Complete ILND + ipsilateral pelvic nodes
L-shaped	Palpable disease; access as S-shaped
Gibson	Ipsilateral pelvic nodes
Midline	Bilateral PLND
T-shaped, elliptical	Bulky nodes; combined inguinal + unilateral pelvic dissection through one incision

Modified from Loughlin, BJU International, 2006.

- **Subinguinal complications:** wound infection, **skin flap necrosis**, lymphocele.

Sequence: oblique skin incision from ASIS toward pubic tubercle, **2 cm below and parallel to the inguinal ligament** (groin crease) → elliptical if disease is palpable or extensive superficially → flaps raised **below Scarpa fascia** → node packet cleared → **myocutaneous flap** closure if resection is extensive.

Concept — *Inguinal incisions exist to reach the cord from above, which is why malignancy mandates the inguinal route. For penile cancer, the node burden, not the primary, chooses the incision.*

INCISIONS FOR SPECIFIC SURGERIES

Scrotal Incisions

- **Elastic skin, easy closure. Only contraindication: presumed testicular or intrascrotal malignancy → inguinal incision.**
- **Unilateral:** transverse or vertical on that side. **Bilateral:** vertical midline through the **median raphe**.
- **Indications:** varicocelectomy, hydrocelectomy, spermatocelectomy, vasectomy, **exploration for torsion**, orchidopexy. Also reaches the penile base for an **inflatable penile prosthesis (IPP)** or **artificial urinary sphincter (AUS)**.
- **Rich blood supply → close dead space** (or leave a temporary **Penrose**) to prevent haematoma. Dartos in one or two layers, absorbable; skin interrupted or horizontal mattress.

Penile Incisions

- Peyronie surgery is covered elsewhere. **Every urologist must manage circumcision, penile fracture and distal shunting for priapism** (shunts: Chapter 70).

Adult circumcision — two-incision “sleeve” (commonest, Fig. 13.13): retract foreskin → mark first circumferential incision **5–10 mm from the glans** (not too close) → reduce prepuce → mark second at the **coronal sulcus** (seen through the prepuce) → recheck both → cut with a **No. 15 blade, guarding the frenular vessels ventrally** (either first; author cuts the outer, preputial one first) → develop plane between prepuce and dartos/shaft → divide collar longitudinally and remove → **meticulous haemostasis** → **3-0 or 4-0 absorbable (chromic gut), simple interrupted**. Continuous closure is possible but risks **strangulation**.

PENILE FRACTURE EXPLORATION — DEGLOVING VS DIRECT INCISION

Degloving	Direct longitudinal
Circumferential subcoronal incision (as in circumcision or Peyronie plication); skin peeled back to expose both corpora	Over the defect; requires pre-operative imaging to localise it
Standard; no localisation needed	Better return of sexual function ; may become the preferred route (Chapter 133)

Perineal Incisions

- Expose the **proximal urethra and penile base**: urethroplasty, perineal urethrostomy, male sling, IPP, AUS; historically **radical perineal prostatectomy**.
- **Urethral surgery: 8–10 cm vertical** incision from the perineoscrotal junction. **Inverted-Y (Lambda) vs midline:** Lambda has **higher 90-day wound complications**.
- **Closure:** layers according to dissection → **Colles fascia closed**, usually continuous, to obliterate dead space → skin in **interrupted mattress** sutures to stop the wound opening in this position.
- **Radical perineal prostatectomy:** large **inverted horseshoe**, apex **~2 cm from the anal verge, ischial tuberosities** as landmarks. Largely replaced by the retropubic route. **Single-port robotic platforms** may revive the perineal route for prostatectomy, cystectomy and PLND.

Use of Retractors

- **Exposure creates the workspace.** **Handheld** (Richardson, Army-Navy) for entry and closure; **self-retaining with padding** to hold the space and protect organs.

RETRACTOR SYSTEMS COMPARED

Retractor	Origin	Design	Classic urological use	Key point
Balfour	1912	Self-retaining, fixed side blades $\pm$ detachable blade	Retropubic (simple/radical prostatectomy), transvesical	Detachable blade retracts bladder; lap pads protect side walls
Finochietto	1941, thoracic surgery	Rib spreader, hand crank	Renal/adrenal via chest approach	Blades between ribs, padded
Bookwalter, Omni-Tract, Thompson	—	Table-fixed full or half ring (wishbone), interchangeable blades	Large incisions, complex cases (cystectomy, nephrectomy with caval thrombus)	Omni-Tract/Thompson extra articulation — no proven benefit
Abdominal wall protector	—	Disposable wound-edge sleeve	Cystectomy, urinary diversion	Fewer wound infections in general surgery; no reduction in urological series
Lone Star	F. Brantley Scott, prosthetic surgery	Ring + elastic hooks	Perineal: prosthetics, urethroplasty	Stepwise retraction layer by layer

- **Table-fixed retractor injury:** bowel, vessels, liver/spleen, nerves. Prevent it by **packing bowel by hand with moist laparotomy pads** first, then applying the **minimum force** needed.

Use of Surgical Drains

- **Purpose:** prevent collections of urine, blood or pus.
- **Passive vs active:** **Penrose (passive)** now mainly in the **scrotum**; **closed suction (Jackson-Pratt, Blake)** in most other surgery.
- **Placement:** separate stab incision, avoiding the **inferior epigastric vessels** in the muscle and the bowel; secured with **nonabsorbable suture**.
- **Little evidence supports routine drains;** routine use is questioned after robotic-assisted radical prostatectomy, partial nephrectomy and robotic cystectomy → **surgeon's discretion**. Drain the renal bed when findings demand it, e.g. **partial nephrectomy, simple nephrectomy for XGP** (Fig. 13.9).

Concept — *Specific-site incisions follow one rule: tissue that is elastic and richly vascular (scrotum, penis, perineum) closes easily but haematomas readily. Dead-space closure and meticulous haemostasis matter more than the incision itself.*

KEY POINTS

- Open approaches still have indications; choice rests on pathology size and complexity, patient preference and surgeon skill.
- Know the abdominal wall anatomy; **staying in the midline avoids key vessels and nerves**.
- Anterior incisions reach pelvic, intraperitoneal, retroperitoneal and extraperitoneal structures; **lower midline is the commonest urological incision** for bladder and prostate.
- The kidney is usually approached through an **anterior or flank** incision, each with trade-offs.
- Inguinal incisions, above or below the ligament, reach the cord, testis and ilioinguinal nodes.

- Adult circumcision: **two-incision sleeve**.
- Scrotal and penile incisions need **meticulous haemostasis**.
- Perineal incisions may return with new robotic technology.